Datos de Voltajes y Amperajes

1. YB-P20M **1**
1.1. OPERATING VOLTAGE: 1
1.2. CURRENT-PER-TORQUE RATIO: 1
1.3. STALL CURRENT (Amperaje máximo): 1
1.4. TORQUE: 1
2. JGB37-520 de Shenzhen Jinshunlaite Motor., Ltd. **2**
2.1. Voltaje: 12 V 2
2.1.1. Sin carga (Amperaje) 2
2.1. Voltaje: 24 V 2
2.1.1. Sin carga (Amperaje) 2
3. ESP32-S3 N16R8 **2**
4. CJMCU-230 SN65HVD230 **3**
4.1. Pines: 3
4.1.1. Vcc (Voltaje suministrado): 3
4.1.3. Digital Input and Output voltage, VI (D or R) (Voltaje): 3
4.1.4. Receiver output current, IO (Amperaje): 3
5. IRF520 PWM **3**
6. IP5306 **4**
7. 7V Voltage Regulator **4**

1. YB-P20M

1.1. OPERATING VOLTAGE:

6–7.4 V

1.2. CURRENT-PER-TORQUE RATIO:

0.0093 A/oz-in (original)
1.317 A / (N *m) (Convertido con IA)
12.915 A / (Kg * m) (Convertido con IA)
0.012915 A / (g * cm) (Convertido con IA)

1.3. STALL CURRENT (Amperaje máximo):

≤3.6 A

1.4. TORQUE:

7.4V 2.7 N-m

2. JGB37-520 de Shenzhen Jinshunlaite Motor., Ltd.

2.1. Voltaje: 12 V

2.1.1. Sin carga (Amperaje)

120 mA

2.1.2. Con carga (Amperaje)

400 A

2.1.3. Parado (Amperaje)

400 A

2.1. Voltaje: 24 V

2.1.1. Sin carga (Amperaje)

50 mA

2.1.2. Con carga (Amperaje)

200 A

2.1.3. Parado (Amperaje)

200 A

3. ESP32-S3 N16R8

Table 6-2. Recommended Operating Conditions

Symbol	Parameter		Min	Typ	Max	Unit
VDD33	Power supply voltage		3.0	3.3	3.6	V
I_{VDD}	Current delivered by external power supply		0.5	—	—	A
T_A	Operating ambient temperature	65 °C version	-40	—	65	°C
		85 °C version			85	
		105 °C version			105	

Table 5-1. Absolute Maximum Ratings

Parameter	Description	Min	Max	Unit
Input power pins[1]	Allowed input voltage	–0.3	3.6	V
I_{output}[2]	Cumulative IO output current	—	1500	mA
T_{STORE}	Storage temperature	–40	150	°C

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4. CJMCU-230 SN65HVD230

4.1. Pines:

4.1.1. Vcc (Voltaje suministrado):

Voltaje: - 0,3V a 6V

4.1.2. CANH or CANL (Voltaje):

Voltaje: -4V a 6V

4.1.3. Digital Input and Output voltage, VI (D or R) (Voltaje):

–0.5V a (VCC + 0.5)V

4.1.4. Receiver output current, IO (Amperaje):

–11mA a 11mA

5. IRF520 PWM

ABSOLUTE MAXIMUM RATINGS (T_C = 25 °C, unless otherwise noted)					
PARAMETER			**SYMBOL**	**LIMIT**	**UNIT**
Drain-source voltage			V_{DS}	100	V
Gate-source voltage			V_{GS}	± 20	
Continuous drain current	V_{GS} at 10 V	T_C = 25 °C	I_D	9.2	A
		T_C = 100 °C		6.5	
Pulsed drain current [a]			I_{DM}	37	
Linear derating factor				0.40	W/°C
Single pulse avalanche energy [b]			E_{AS}	200	mJ
Repetitive avalanche current [a]			I_{AR}	9.2	A
Repetitive avalanche energy [a]			E_{AR}	6.0	mJ
Maximum power dissipation	T_C = 25 °C		P_D	60	W
Peak diode recovery dV/dt [c]			dV/dt	5.5	V/ns
Operating junction and storage temperature range			T_J, T_{stg}	-55 to +175	°C
Soldering recommendations (peak temperature) [d]	For 10 s			300	
Mounting torque	6-32 or M3 screw			10	lbf · in
				1.1	N · m

Notes

a. Repetitive rating; pulse width limited by maximum junction temperature (see fig. 11)
b. V_{DD} = 25 V, starting T_J = 25 °C, L = 3.5 mH, R_g = 25 Ω, I_{AS} = 9.2 A (see fig. 12)
c. $I_{SD} \leq 9.2$ A, dI/dt ≤ 110 A/µs, $V_{DD} \leq V_{DS}$, $T_J \leq 175$ °C
d. 1.6 mm from case

6. IP5306

6 Limit parameters

parameter	symbol	value	unit
Port input voltage range	V_{IN}	-0.3 to 5.5	V
Operating temperature range	T_A	0 ~ 70	℃
Junction Temperature Range	T_J	- 40 to 150	℃
Storage Temperature Range	Tstg	- 60 to 150	℃
Thermal resistance (junction to ambient)	θ_{JA}	50	℃ / W
Human Body Model (HBM)	ESD	4	KV

7. 7V Voltage Regulator

Duda, tiene que ser de entrada a 7 y salida entre 12 y 24 para los servors o al revés para que el esp32 pueda recibir la señal de los servos?